... the lowest program execution?

a. Pure interpreter
b. Compiler
c. Hybrid interpreter
d. Virtual machine

3) Aliasing is harmful for:
a. Writability
b. Flexibility
c. Readability and reliability
d. none

* Reliability

4) Design issues of variables names includes:
a. Case sensitivity
b. Name length
c. Special words
d. All of the above

5) The lifetime of local variable in ***Java*** language is:
a. Static
b. Explicit heap-dynamic
c. Stack-dynamic
d. none of the above

6) (T / F) The heap is a collection of storage cells which is disorganized.

7) (T / F) The garbage collection attempts to get back memory occupied by objects that are no longer accessible to the user program.

8) (T / F) It is possible to have multiple variables with the same address.

9) (T / F) It is possible to have a variable with different addresses in different times.

10) Consider the following grammar to define identifiers

$b1234

```
<identifier> →  $ <letter> {<letter>|<digit>}
<letter>     →  a|b|c|d
<digit>      →  1|2|3|4|5|6|7|8|9|0
```

<iden> → $<
→ $
→ $

Which of the following identifiers are accepted by the grammar? (Yes/No)

CS301 Programming Languages First Exam: 2nd Semester 1434h Time: 80 Mi

me: ID: Section: CAR

5 Points] **Question 1**: Choose the **correct answer** for each of the following statements:

1) Consider this Java statement: $x = 2 * y + y$; which of the following is a token:

a. x
b. +
c. y
d. semicolon

2) Which produces the lowest program execution?

a. Pure interpreter
b. Compiler
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2. What is the terminal set?

The terminal is $\{+, -, (\ ), i, c\}$

3. Determine the attribute type in each rule (beside each rule)

1- E.actual type = E[1].actual type + E[2].actual type

4. What is the original context free grammar? (Write down the CFG)

$E \rightarrow E[1]\ (+ \mid -)\ E[2]$
$\mid -E[1]$
$\mid (E[1])$
$\mid (i, c)$

$\boxed{\text{The BNF}}$

$E \rightarrow E[1] + E[2]$
$\mid E[1] - E[2]$
$\mid -E[1]$
$\mid (E[1])$
$\mid i$
$\mid c$

5. What are the benefits of attribute grammar over context free grammar?

The "conten free grammer" give a describtion of the syntax structure of the language But the "Attribute Grammer" give the descrition of the syntax and the ~~indirect~~ meaning of it ((semantic of each
mple, you can determine and check the Type compatibility using
Grammer But you can't do that using "conten-free Gramme
g an attribute grammer, a "fully attribute Parse Tree" can be get
tion, Two types of Attributes can be determines By Att
which

→ Capital Letter are Non-terminal
→

actual type	synthize attribute
expected type	inherit attribute

[3 Points] **Question 6**: Given the following attribute grammar

1. E→E[1]+E[2] {Semantic Rule: E.val:= E[1].val+E[2].val} [1] synthize attribute "actual type"
2. E→E[1]-E[2] {Semantic Rule: E.val:= E[1].val-E[2].val} [2] synthize attribute = actual type
3. E→-E[1]{Semantic Rule: E.val:= -E[1].val} [3] synthize attribute = actual type
4. E→(E[1]) {Semantic Rule: E.val:= E[1].val} [4] synthize attribute = actual type
5. E→i {Semantic Rule: E.val:=symbTable(i.name,type)} [5] synthize attribute = actual type
6. E→c {Semantic Rule: E.val:= c.val} [6] synthize attribute → actual type

Answer the following question:

1) What is the non-terminal set?
The Non Terminal is { E }

2) What is the terminal set?
The terminal is { + , - , ( ) , i , c }

3) Determine the attribute type in each rule (beside each rule)
1- E.actual type = E[1].actual type + E[2].actual type

4) What is the original context free grammar? (Write down the CFG)

E → E[1] (+ | -) E[2]
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5) What are the benefits of attribute grammar over context free grammar?

The "Context free grammer" give a description of the syntax structure of the language. But the "Attribute Grammer" give the describtion of the syntax and the indirect meaning of it (semantic of each

1. (Yes) $b1234
2. (Yes) $abcd1234
3. (No) $$d1234
4. (No) d1234
5. (No) $L1234
6. (No) $bbcc$

[2 Points] **Question 2**: Compute the weakest precondition for the following statement and postconditions (**show your answer**):

statement1 a = 3 * b - 1;
Statement2 b = a - 3 {b > -5}

$\Rightarrow$ $\{b > \frac{-1}{3}\}$ a = 3*b - 1 $\{a > -2\}$
$\{a > -2\}$ b = a - 3 $\{b > -5\}$

$\boxed{1}$

$b > -5$
$a - 3 > -5$
$a > -5 + 3$
$\{a > -2\} \Rightarrow$ postcondition of statement1 and precondition of statement2

$\boxed{2}$

$a > -2$
$3*b - 1 > -2$
$3*b > -2 + 1$
$\{b > \frac{-1}{3}\} \Rightarrow$ precondition of state… precondition of both s…

[2 Points] **Question 3**: What are the output of lexical analyzer phase? And what is the usage of symbol table?

lexical analyzer → output is Lexemes

Symbol table is use to store the Variables type and Names, and … ~~the value~~ of them when reach to the leaf of a parse tree, which … the output … Syntax …

[1 Points] **Question 4**: Justify, **Java programming language** has $\boxed{\text{recursion}}$ mechanism?

Java programing language has a Stack-Dynamic Binding Variables (…
This kind of variable are allowed the Recursion of the Sub…
since it is allow sharing the memory location between …
So it keeps an active copy of the variables to each …

[2 Points] Question 5 …

1. (Yes) \$b1234
2. (Yes) \$abcd1234
3. (No) \$\$d1234
4. (No) d1234
5. (No) \$L1234
6. (No) \$bbcc\$

[2 Points] **Question 2**: Compute the weakest precondition for the following statement and postconditions (**show your answer**):

Statment 1 `a = 3 * b - 1;`

Statment 2 `b = a - 3 {b > -5}`

$\{b > \frac{-1}{3}\}$ a = 3*b - 1 $\{a > -2\}$

$\{a > -2\}$ b = a - 3 $\{b > -5\}$

$\boxed{1}$

$b > -5$

$a - 3 > -5$

$a > -5 + 3$

$\{a > -2\}$ ⇒ postcondition of statment 1 and precondition of stament 2

$\boxed{2}$

$a > -2$

$3 * b - 1 > -2$

$3 * b > -2 + 1$

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[2 Points] **Question 3**: What are the output of lexical analyzer phase? And what is the usage of symbol table?

lexical analyzer → output is Lexemes

Symbol table is use to store the Variables type and names, and to get ~~the value~~ of them when reach to the leaf of a parse tree which is the output of the syntax analyzer

[1 Points] **Question 4**: Justify, **Java programming language** has recursion mechanism?

Java programming language has a Stack-Dynamic Binding Variables (Local Variables). This kind of variable are allowed the Recursion of the Subprograms since it is allow sharing the memory location between Recursive subprograms So it keeps an active copy of the variables to each Recursion subprograms.

[2 Points] **Question 5**: What are the differences between parse tree and derivations?

Parse Tree: 1) give an ability to check ambiguity

Two distict parse tree for the same input means, there is ambiguity in language grammar.

2) also to check the priority of operators in the given statment base on the grammar, which operators prece[illegible] another

3) it give an easier implementation of programme and in excution time

4) Parse Tree is the output of the syntax analyzer in compil[illegible] process (also in hybrid-implementation system), which ma[illegible] the process more easy to the semantic analyzer convert it into intermediat code.

Name: ... Exam: 2nd Semester 1434h Time: 60 Min.

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[5 Points] **Question 1**: Choose the **correct answer** for each of the following statements:

1) Consider this Java statement: $x = 2 * y + y$; which of the following is a token:
   a. x
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   d. semicolon

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$b123

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